

Trainings ▾

Preparatory Steps

- Remove the gear cover. Refer to Procedure 001-031 in Section 1. (</qs3/pubsys2/xml/en/procedures/35/35-001-031-tr.html>)

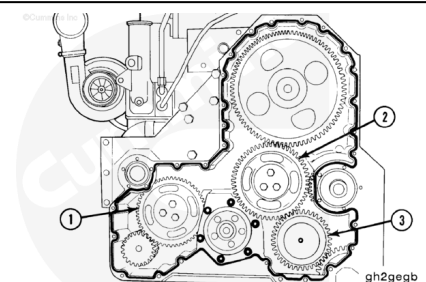


Remove

Three idler gear assemblies are used:

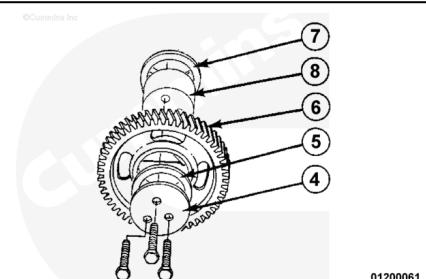
- Water pump idler gear (1)
- Camshaft idler gear (2)
- Hydraulic pump idler gear (3).

The water pump **must** be removed before the water pump idler gear (1) can be removed. Refer to Procedure 008-062 in Section 8. (</qs3/pubsys2/xml/en/procedures/35/35-008-062-tr.html>)

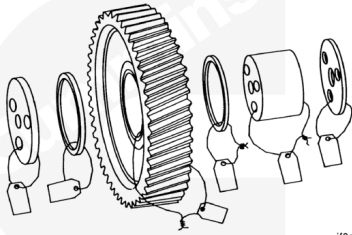


To remove the water pump idler gear assembly, remove:

- The three capscrews
- Cover plate (4)
- Front thrust bearing (5)
- Idler gear (6)
- Rear thrust bearing (7)
- Idler gear shaft (8).



Mark or tag the individual pieces of the idler gear assembly and attach them together.

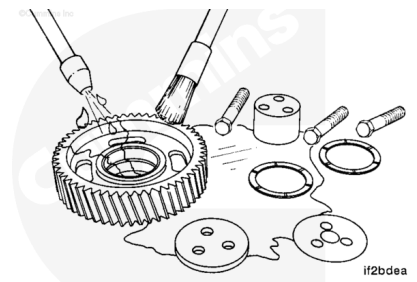


if2geba

Clean and Inspect for Reuse

⚠ WARNING ⚠

When using solvents or alkaline materials for cleaning, follow the manufacturer's recommendations for use. Wear goggles and protective clothing to reduce the possibility of personal injury.



if2bdea

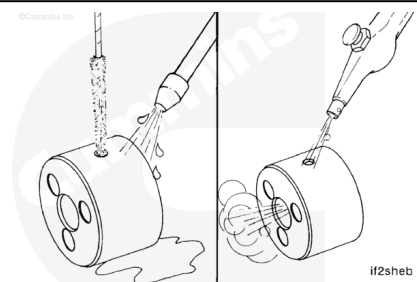
⚠ WARNING ⚠

Wear appropriate eye and face protection when using compressed air. Flying dirt and debris can cause personal injury.

Clean the parts with solvent. Dry with compressed air.

⚠ WARNING ⚠

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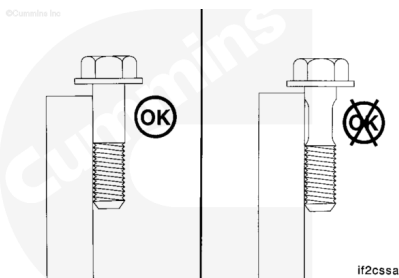


if2sheb

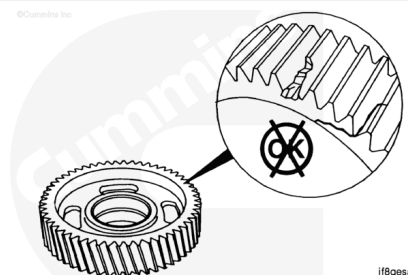
Use a bristle brush to clean the oil drilling in the idler gear shaft.

Blow out the oil drilling with compressed air.

Use a straightedge to inspect the retaining capscrews for necking. If necking is observed, do **not** reuse the capscrews. They **must** be replaced.



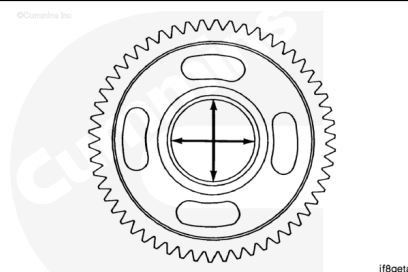
Inspect the idler gear for chipped, broken, or cracked teeth.



Measure the inside diameter of the idler gear bushing bore.

Water Pump Idler Gear Bushing Bore Inside Diameter

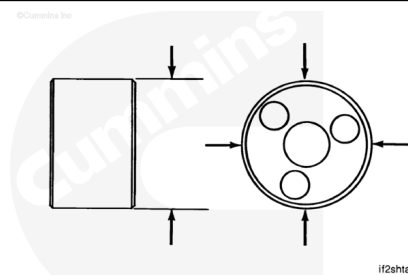
mm		in
60.045	MIN	2.3640
60.100	MAX	2.3661



Measure the outside diameter of the idler gear shaft.

Water Pump Idler Gear Shaft Outside Diameter

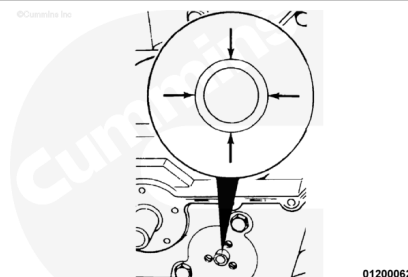
mm		in
59.975	MIN	2.3612
60.008	MAX	2.3625



Measure the outside diameter of the ring dowel.

Water Pump Idler Ring Dowel Outside Diameter

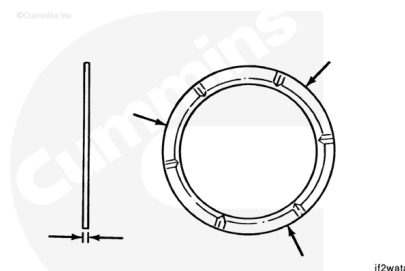
mm		in
19.217	MIN	0.7566
19.243	MAX	0.7576



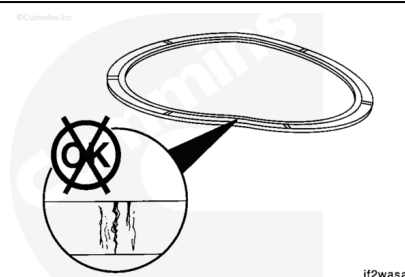
Measure the thrust washer thickness in three places 120-degrees apart.

Water Pump Idler Thrust Washer Thickness

mm		in
2.400	MIN	0.0945
2.470	MAX	0.0972



Flex the thrust washers approximately 3 to 6 mm [1/8 to 1/4 in], and inspect the surfaces for cracks.



Install

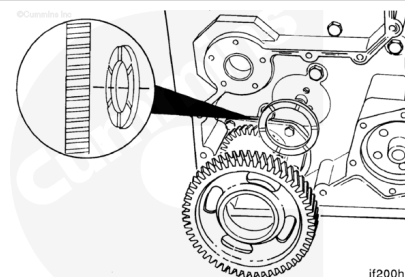
⚠ CAUTION ⚠

The grooved side of the rear thrust bearing must be facing toward the gear to prevent damage to the gear and engine during engine operation.

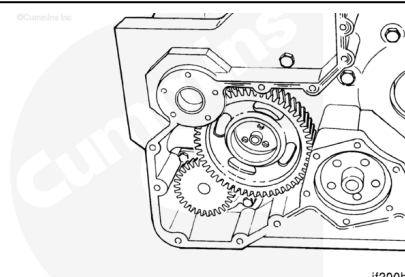
Use Lubriplate™ 105, or equivalent, to lubricate the thrust bearings and idler gear shaft.

Install the rear thrust bearing.

Install the idler gear without the shaft.



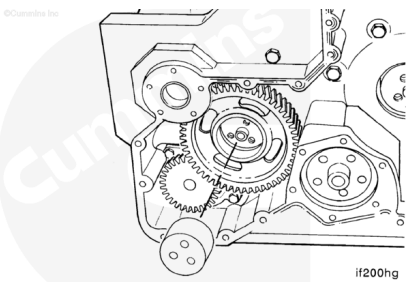
Align the inside diameter of the rear thrust bearing with the inside diameter of the idler gear.



The shaft **must** pass through the gear and pilot into the rear thrust bearing. After installation, the shaft **must** protrude **only** slightly more than the thickness of the front thrust bearing.

Excessive shaft protrusion beyond the thickness of the front thrust bearing indicates the shaft is **not** properly piloted into the rear thrust bearing.

Install the shaft into the gear bore.



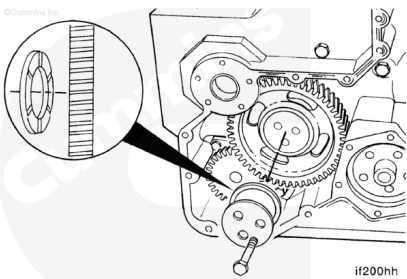
⚠ CAUTION ⚠

The grooved side of the front thrust bearing must be facing toward the gear to prevent damage to the gear and engine during engine operation.

Install the front thrust bearing.

Install the gear retainer.

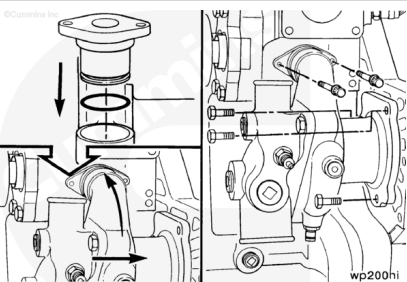
Install the capscrews and tighten.



Torque Value:

1. 61 n•m [45 ft-lb]
2. Rotate 60 degrees

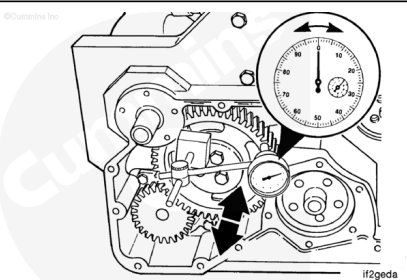
Install the water pump. Refer to Procedure 008-062 in Section 8. (</qs3/pubsys2/xml/en/procedures/35/35-008-062-tr.html>)



The following idler gear end clearance and backlash checks apply to all three idler gears. The same MINIMUM and MAXIMUM values apply, also.

Use a dial indicator gauge with a magnetic base to measure the idler gear end clearance.

Place the contact tip of the gauge against the face of the idler gear.



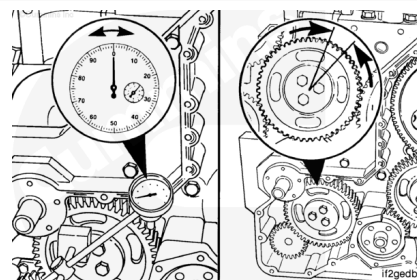
Idler Gear End Clearance

mm		in
0.30	MIN	0.012
0.53	MAX	0.021

Use a dial indicator gauge with a magnetic base to measure the idler gear backlash.

Place the contact tip of the gauge against a tooth on the idler gear.

Do **not** allow the mating gears to move while measuring the backlash.



Idler Gear Backlash

mm		in
0.08	MIN	0.003
0.38	MAX	0.015

Finishing Steps

- Install the gear cover. Refer to Procedure 001-031 in Section 1. (</qs3/pubsys2/xml/en/procedures/35/35-001-031-tr.html>)
- Operate the engine to normal operating temperature and check for leaks.

